

Sapphire's Toxic 5850

Complete with custom PCB, chokes, caps and cooling, Sapphire's Toxic HD 5850 promises more of the Cypress goodness we've got a taste of

How stuff works

of AMD SPs to NVIDIA SPs is something like 5:1. Therefore, for a practical comparison, the HD 5870 can be said to have 1600/5 or 320 SPs, while NVIDIA GTX 285 has 240. Those who are familiar with comparative benchmarks will agree, for despite the boast of 1600 SPs, the HD 5870 is not five times faster than the GTX 285. The GTX 480 has 480 SPs, though we won't know for sure till after the launch thanks to NVIDIA's secrecy. To feed the extra SPs, the HD 5870 has double the L2 cache size — 128 KB, up from 64 KB on the HD 4870.

There's also a doubling of texture units from the HD 4870, the count is now at 80. The texture units are now right next to the thread processors instead of being linked, so if AMD adds more thread processors, the texture units

digit

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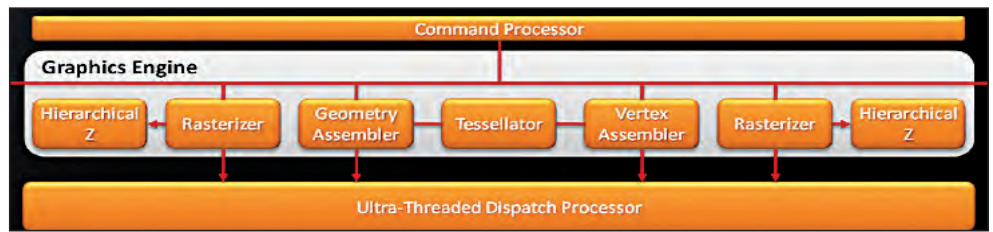
will also go up. For every SIMD engine, there are four texture units.

The graphics engine

The HD 4870 had a Tessellator, but this cursory addition has seen a major overhaul with the addition of Hull and Domain Shaders giving it full DX11 compliance. AMD has also added an algorithm to reduce tessellation artifacts and there are two Tessellator units instead of one. There are also two Raster units, obviously necessary to feed double the SPs.

USB 3.0 in demand

NEC plans to double their production of USB 3.0 controllers due to the high demand. Hopefully that means we'll see more of it on value motherboards



This is the part AMD calls the Graphics Engine. It consists of Tessellator and Raster Units among others

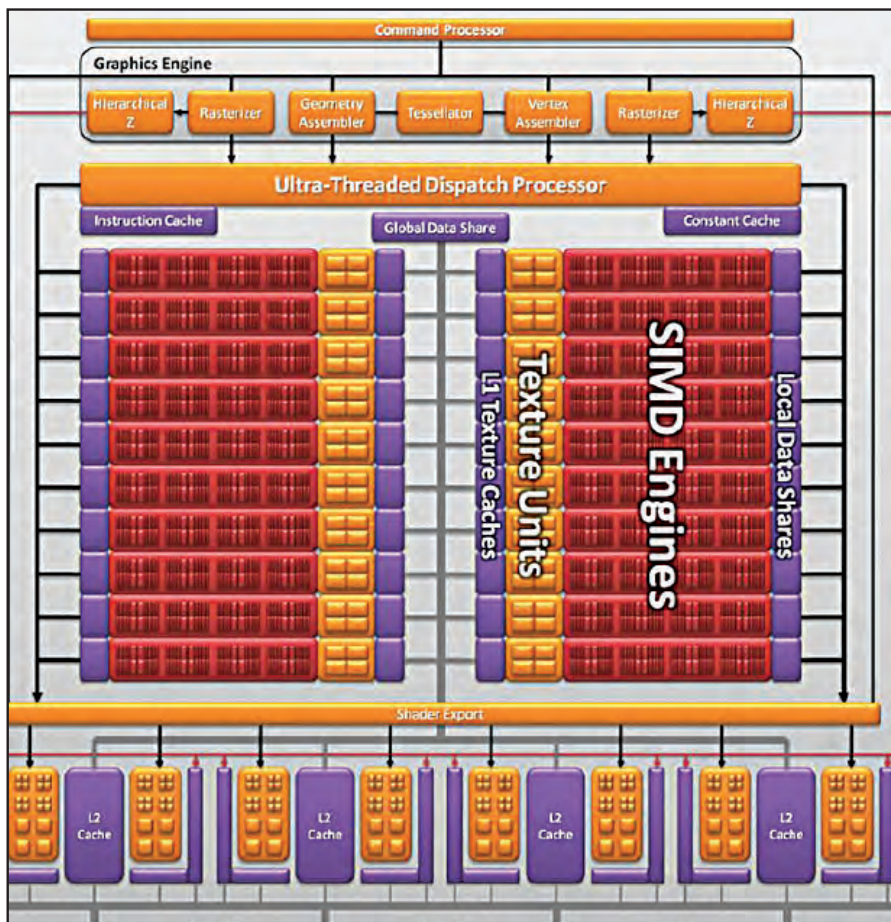
Among other DX11-specific instructions that have been added, a new instruction called SAD or sum of absolute differences is also supported. This is of great importance in (among other applications) video encoding. This instruction is used in motion estimation between adjacent frames and previously emulation of this function required 12 clock cycles; now this is down to one cycle — resulting in a 2x boost to video encoding speed.

Benefits all round

DX11 and a compliant card is a good thing to have for most users. Tessellation, along with the Hull Shader and Domain Shader translate to improved image quality via injection of extra details in a scene at a minimal performance hit. NVIDIA has also taken Tessellation seriously with their GF100-based GPUs, but AMD gets a victor point here for being the pioneer and pushing the technology even before it was a part of the Direct X spec-sheet.

The Compute Shader is another notable addition. This allows programs to access the GPU and treat it like a generic processor rather than a graphics processor. This is what we know as the *GPGPU* concept and while we haven't gotten there fully, this is a notable step forward. The Compute Shader is used by games and non-game applications and is also known as DirectCompute, so do not let the jargon confuse you.

One of the more exciting features that AMD is really pushing is OIT or Order Independent Transparency. This will really make gamers happy. The example of OIT benefitting gamers that Richard Huddy gave us was the result of feedback from a major game developer. They wanted a way to be able to properly render a transparent dragon crossing a river and breathing fire on to



The block diagram for the Radeon HD 5870